

Foundations of Explainable artificial intelligence

[Explainable artificial intelligence]

1.0 Content Block

Adaptive integration and explanation Many approaches that it uses provides explanation in general, it doesn't take account for the diverse backgrounds and knowledge level of the users. This leads to challenges with accurate comprehension for all users. Expert users can find the explanations lacking in depth, and are oversimplified, while a beginner user may struggle understanding the explanations as they are complex. This limitation downplays the ability of the XAI techniques to appeal to their users with different levels of knowledge, which can impact the trust from users and who uses it. The quality of explanations can be different amongst their users as they all have different expertise levels, including different situation and conditions.

2.0 Content Block

Understanding versus trust The goal of explainability to end users of AI systems is to increase trust in the systems, even "address concerns about lack of 'fairness' and discriminatory effects". However, even with a good understanding of an AI system, end users may not necessarily trust the system. In one study, participants were presented with combinations of white-box and black-box explanations, and static and interactive explanations of AI systems. While these explanations served to increase both their self-reported and objective understanding, it had no impact on their level of trust, which remained skeptical. This outcome was especially true for decisions that impacted the end user in a significant way, such as graduate school admissions. Participants judged algorithms to be too inflexible and unforgiving in comparison to human decision-makers; instead of rigidly adhering to a set of rules, humans are able to consider exceptional cases as well as appeals to their initial decision. For such decisions, explainability will not necessarily cause end users to accept the use of decision-making algorithms. We will need to either turn to another method to increase trust and acceptance of decision-making algorithms, or question the need to rely solely on AI for such impactful decisions in the first place. However, some emphasize that the purpose of explainability of artificial intelligence is not to merely increase users' trust in the system's decisions, but to calibrate the users' level of trust to the correct level. According to this principle, too much or too little user trust in the AI system will harm the overall performance of the human-system unit. When the trust is excessive, the users are not critical of possible mistakes of the system and when the users do not have enough trust in the system, they will not exhaust the benefits inherent in it.

3.0 Content Block

Within artificial intelligence (AI), explainable AI (XAI), generally overlapping with interpretable AI or explainable machine learning (XML), is a field of research that explores methods that provide humans with the ability of intellectual oversight over AI algorithms. The main focus is on the reasoning behind the decisions or predictions made by the AI algorithms, to make them more understandable and transparent. This addresses users' requirement to assess safety and scrutinize the automated decision making in applications. XAI counters the "black box" tendency of machine learning, where even the AI's designers cannot explain why it arrived at a specific decision. XAI hopes to help users of AI-powered systems perform more effectively by improving their understanding of how those systems reason. XAI may be an implementation of the social right to explanation. Even if there is no such legal right or regulatory requirement, XAI can improve the user experience of a product or service by helping end users trust that the AI is making good decisions. XAI aims to explain what has been done, what is being done, and what will be done next, and to unveil which information these actions are based on. This makes it possible to confirm existing knowledge, challenge existing knowledge, and generate new assumptions.

4.0 Content Block

Scholars sometimes use the term "mechanistic interpretability" to refer to the process of reverse-engineering artificial neural networks to understand their internal decision-making mechanisms and components, similar to how one might analyze a complex machine or computer program. Studying the interpretability of the most advanced foundation models often involves searching for an automated way to identify "features" in generative pretrained transformers. In a neural network, a feature is a pattern of neuron activations that corresponds to a concept. A compute-intensive technique called "dictionary learning" makes it possible to identify features to some degree. Enhancing the ability to identify and edit features is expected to significantly improve the safety of frontier AI models. For convolutional neural networks, DeepDream can generate images that strongly activate a particular neuron, providing a visual hint about what the neuron is trained to identify.

5.0 Content Block

Payoff allocation Nizri, Azaria and Hazon present an algorithm for computing explanations for the Shapley value. Given a coalitional game, their algorithm decomposes it to sub-games, for which it is easy to generate verbal explanations based on the axioms characterizing the Shapley value. The payoff allocation for each sub-game is perceived as fair, so the Shapley-based payoff allocation for the given game should seem fair as well. An experiment with 210 human subjects shows that, with their automatically generated explanations, subjects perceive Shapley-based payoff allocation as significantly fairer than with a general standard explanation.

6.0 Content Block

Participatory budgeting Yang, Hausladen, Peters, Pournaras, Fricker and Helbing present an empirical study of explainability in participatory budgeting. They compared the greedy and the equal shares rules, and three types of explanations: mechanism explanation (a general explanation of how the aggregation rule works given the voting input), individual explanation (explaining how many voters had at least one approved project, at least 10000 CHF in approved projects), and group explanation (explaining how the budget is distributed among the districts and topics). They compared the perceived trustworthiness and fairness of greedy and equal shares, before and after the explanations. They found out that, for MES, mechanism explanation yields the highest increase in perceived fairness and trustworthiness; the second-highest was Group explanation. For Greedy, Mechanism explanation increases perceived trustworthiness but not fairness, whereas Individual explanation increases both perceived fairness and trustworthiness. Group explanation decreases the perceived fairness and trustworthiness.

7.0 Content Block

Background Machine learning (ML) algorithms used in AI can be categorized as white-box or black-box. White-box models provide results that are understandable to experts in the domain. Black-box models, on the other hand, are extremely hard to explain and may not be understood even by domain experts. XAI algorithms follow the three principles of transparency, interpretability, and explainability.